

1. Project Description

The **Clinic Management System** is a comprehensive digital platform designed to automate healthcare workflows. It serves as a central hub connecting Patients, Doctors, Receptionists, and Administrators. By digitizing medical records and scheduling, the system minimizes human error, improves patient care quality, and provides a minimalist, user-friendly experience for all clinical staff.

2. System Actors

A. Patient

The patient is the primary end-user who interacts with the system to manage their healthcare journey.

- **Authentication:** Users can securely login and logout of their personal dashboard.
- **Account Management:** New patients can register an account, while existing users can update their personal profiles.
- **Appointment Handling:** Patients have the authority to book new appointments, cancel existing ones, or reschedule based on availability.
- **Accessing Records:** Patients can view their digital prescriptions and check their upcoming or past appointments.

B. Receptionist

The receptionist manages the daily flow of patients and initial administrative tasks.

- **Patient Intake:** Manages walk-in registration by adding new patients directly to the system.
- **Schedule Oversight:** Manages appointment lists and performs the Check-in process when a patient arrives.
- **Status Updates:** Responsible for updating the status of appointments (confirmed, pending, completed).

C. Doctor

Doctors use the system to manage clinical data and patient health outcomes.

- **Patient Review:** Accesses and views patient medical history and current appointment schedules.
- **Clinical Assessment:** Records medical diagnoses and requests necessary laboratory tests.

- **Digital Prescriptions:** Writes and issues prescriptions, which includes the ability to add specific medications and link them to a diagnosis.

D. Admin

The admin ensures the system runs smoothly and handles high-level configurations.

- **System Control:** Manages all system users and monitors the overall clinic schedule.
- **Staff Coordination:** Authorized to add new doctors to the clinic or remove them as needed.
- **Analytics:** Generates comprehensive clinical reports and views statistics to monitor clinic performance.

E. Bank System

An external financial interface that handles all money transactions.

- **Transaction Processing:** Verifies and completes payments for appointment bookings.
- **Financial Reversals:** Triggers and processes refunds automatically when an appointment is canceled.

3. Functional Requirements

1 Authentication & Security

- The system shall provide a secure login and logout mechanism for all identified actors.
- The system shall enforce role-based access control to ensure users only access features relevant to their role.

2 Patient & Profile Management

- The system shall support self-registration for new patients.
- The system shall allow users to edit and update their profile information at any time.
- The system shall allow the receptionist to manually register patients who arrive at the clinic without an account.

3 Appointment & Scheduling

- The system shall facilitate appointment booking, which requires a successful payment verification from the Bank System.
- The system shall allow for appointment rescheduling and cancellations.
- Cancellation of an appointment shall automatically trigger a refund process through the external Bank System.

- The receptionist shall have the capability to check-in patients and update appointment statuses in real-time.

4 Clinical Operations

- The system shall allow doctors to view a comprehensive medical history for any assigned patient.
- The system shall enable doctors to record diagnoses and link them to digital prescriptions.
- The system shall support the addition of multiple medications within a single prescription record.

5 Administration & Reporting

- The system shall allow the Admin to manage the database of doctors (adding/removing staff).
- The system shall provide tools to view clinic statistics and generate performance reports.

4. Non-Functional Requirements

- **Usability:** The interface shall be minimalist and intuitive, ensuring that staff can perform core tasks with less than 2 hours of training.
- **Performance:** The system shall respond to data queries and load medical records in less than 2 seconds.
- **Integrity:** Sensitive medical data shall be encrypted and only accessible to the assigned Doctor and Administrator.
- **Reliability:** The system shall maintain 99.9% uptime during the clinic's operating hours.
- **Portability:** The application shall be compatible with all modern web browsers and responsive across various devices.